

We Love Beamer

Every overlay, block and column construct, annotated

Zarif Ishmam

Department of Computer Science and Engineering, BUET

September 21, 2026

Outline

- 1 Frames and overlays
- 2 Columns, tables and figures
- 3 Blocks and theorems
- 4 Frame options

I am the TITLE

I am the FRAME.

I am the TITLE

I am the FRAME. I build up beamer presentations.

I am the TITLE

I am the FRAME. I build up beamer presentations. It is paused.

The other way to set a title

and a subtitle underneath it

Same result, different syntax.

Lists that build up

- Point 1
- Point 2
 - part 2.1
 - part 2.3
- You are wondering where part 2.2 is
- Mind reading!

Lists that build up

- Point 1
- Point 2
 - part 2.1
 - part 2.3
- You are wondering where part 2.2 is
- Mind reading!

Lists that build up

- Point 1
- Point 2
 - part 2.1
 - part 2.3
- You are wondering where part 2.2 is
- Mind reading!

Lists that build up

- Point 1
- Point 2
 - part 2.1
 - part 2.3
- You are wondering where part 2.2 is
- Mind reading!

Explicit overlay numbers

- ❶ Look at part 1
- ❷ Look at I
 - Look at part 2
 - Look at III
- ❸ Nothing to see here, on to the next frame ...
- ❹ Look at II

Explicit overlay numbers

- ❶ Look at part 1
- ❷ Look at I
 - Look at part 2
 - Look at III
- ❸ Nothing to see here, on to the next frame ...
- ❹ Look at II

Explicit overlay numbers

- ❶ Look at part 1
- ❷ Look at I
 - Look at part 2
 - Look at III
- ❸ Nothing to see here, on to the next frame ...
- ❹ Look at II

Explicit overlay numbers

- ❶ Look at part 1
- ❷ Look at I
 - Look at part 2
 - Look at III
- ❸ Nothing to see here, on to the next frame ...
- ❹ Look at II

Explicit overlay numbers

- ❶ Look at part 1
- ❷ Look at I
 - Look at part 2
 - Look at III
- ❸ Nothing to see here, on to the next frame ...
- ❹ Look at II

only vs uncover vs visible

`\uncover` – space reserved, dimmed

`\alt` showing the FIRST slide's text

Auto-incrementing, so you never count slides by hand:

- first click
- second click
- third click

only vs uncover vs visible

`\only` – no space reserved
`\uncover` – space reserved, dimmed
`\visible` – space reserved, not dimmed
`\alt` showing the SECOND slide's text

Auto-incrementing, so you never count slides by hand:

- first click
- second click
- third click

only vs uncover vs visible

`\only` – no space reserved

`\uncover` – space reserved, dimmed

`\visible` – space reserved, not dimmed

`\alt` showing the FIRST slide's text

Auto-incrementing, so you never count slides by hand:

- first click
- second click
- third click

Text formatting with overlays

Mission Hexa

Mission Hexa

Mission Hexa

Mission Hexa

Mission Hexa

Mission Hexa

Mission Hexa

Mission Hexa

Text formatting with overlays

Mission Hexa

Mission Hexa

Mission Hexa

Mission Hexa

Mission Hexa

Mission Hexa

Mission Hexa

Mission Hexa

Text formatting with overlays

Mission Hexa

Mission Hexa

Mission Hexa

Mission Hexa

Mission Hexa

Mission Hexa

Mission Hexa

Mission Hexa

Description lists

API Application Programming Interface

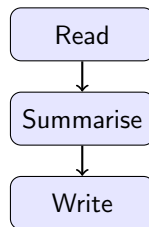
LAN Local Area Network

ASCII American Standard Code for Information Interchange

Using columns

The switch form and the environment form are interchangeable. Everything you would put on a slide can go in a column: lists, maths, tables, even a tikzpicture.

- one
- two



A table revealed row by row

Competitor Name	Swim	Cycle	Run	Total
John T	13:04	24:15	18:34	55:53
Norman P	8:00	22:45	23:02	53:47
Alex K	14:00	28:00	n/a	n/a
Sarah H	9:22	21:10	24:03	54:35

Table: Triathlon results

A table revealed row by row

Competitor Name	Swim	Cycle	Run	Total
John T	13:04	24:15	18:34	55:53
Norman P	8:00	22:45	23:02	53:47
Alex K	14:00	28:00	n/a	n/a
Sarah H	9:22	21:10	24:03	54:35

Table: Triathlon results

A table revealed row by row

Competitor Name	Swim	Cycle	Run	Total
John T	13:04	24:15	18:34	55:53
Norman P	8:00	22:45	23:02	53:47
Alex K	14:00	28:00	n/a	n/a
Sarah H	9:22	21:10	24:03	54:35

Table: Triathlon results

A table revealed row by row

Competitor Name	Swim	Cycle	Run	Total
John T	13:04	24:15	18:34	55:53
Norman P	8:00	22:45	23:02	53:47
Alex K	14:00	28:00	n/a	n/a
Sarah H	9:22	21:10	24:03	54:35

Table: Triathlon results

The three block environments

Real Madrid

Greatest football club in the world.

Surprising

Barcelona conceded 8 goals in one leg!

UCL Titles

RMA has 15 titles!

Theorem-like environments are pre-defined

Theorem (Pythagoras)

$$a^2 + b^2 = c^2$$

Corollary

$$x + y = y + x$$

Proof.

$$\omega + \phi = \epsilon$$



Theorem-like environments are pre-defined

Theorem (Pythagoras)

$$a^2 + b^2 = c^2$$

Corollary

$$x + y = y + x$$

Proof.

$$\omega + \phi = \epsilon$$



Theorem-like environments are pre-defined

Theorem (Pythagoras)

$$a^2 + b^2 = c^2$$

Corollary

$$x + y = y + x$$

Proof.

$$\omega + \phi = \epsilon$$



Example (a theorem-like environment)

Its argument is an optional NAME in brackets.

a coloured box

Its argument is a required TITLE in braces.

Definition

A definition.

Code needs [fragile]

```
def writhe(signs):  
    return sum(signs)
```

Without [fragile] this frame does not compile.

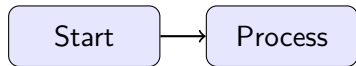
[shrink=10] rescues an overflowing frame

If the content is slightly too tall, `shrink` scales it down rather than letting it run off the bottom. `allowframebreaks` is the alternative: it spills the frame onto extra slides and appends a roman numeral to the title.

[plain] hides the headline and footline

Useful for a full-bleed image or a section divider.

TikZ in a frame, staged with `\only`



TikZ in a frame, staged with `\only`



Cheat sheet

- `\pause` – everything after waits one click
- `<3->` `<-3>` `<2-4>` `<1,3,5>` `<+>` – overlay specifications
- `\only` no space | `\uncover` / `\visible` keep space
- `\onslide<2->` – a switch, reset it afterwards
- `block` / `alertblock` / `exampleblock`
- `[fragile]` for code, `[shrink=10]` for overflow, `[plain]` for full bleed
- `\setbeamercovered{transparent}` – dim, do not hide
- `\setbeamertemplate{navigation symbols}{}` – kill the arrows